

Stress-Testing Bangladesh's Roads for Disaster Readiness – An Agent-Based Network Simulation

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Executive summary

- This report documents simulation outcomes from a data-driven traffic disruption model of Bangladesh's road network, focusing on the national highways N1 and N2 and their connecting roads.
- Bridge disruptions are represented as stochastic breakdown events that induce delays on affected network components. Scenarios progressively increase breakdown probabilities for lower-condition bridge categories.
- Across scenarios, average delay time increases moderately, but total driving time increases sharply (approximately six-fold from baseline to the most severe scenario), indicating strong sensitivity to where disruptions occur in the network.
- Network-analytic metrics (betweenness and closeness centrality) consistently identify a small set of bridges and intersections - largely located on N1 - as critical for system performance.
- Policy implication: resilience investments should prioritize (i) high-centrality bridges and junctions, and (ii) assets where poor condition coincides with high network centrality, as these yield the largest expected reduction in disruption-driven travel time.

1. Policy context and analytical question

Bangladesh's transport system operates under high exposure to climate-related hazards and recurrent disruptions. For policy-makers, a practical question is where limited maintenance and adaptation budgets should be spent to maximize network resilience. This analysis uses simulation and network metrics to identify road network components whose disruption most affects travel time, and to derive a prioritization logic for resilience investments.

2. Data, preprocessing, and network construction

The road and bridge datasets were cleaned to remove obvious geometric and semantic inconsistencies (e.g., outlier road points, implausible bridge locations) and transformed into a graph representation. Nodes represent intersections and bridge locations; edges represent road segments. Additional intersection points were inferred where road geometries overlap but were not explicitly labelled.

2.1 Data quality improvements used in the simulation

Figure 1 illustrates the removal and correction of outlier road geometries during preprocessing. This step is important because geometric outliers can create artificial shortcuts or disconnected components, biasing route choice and centrality metrics.

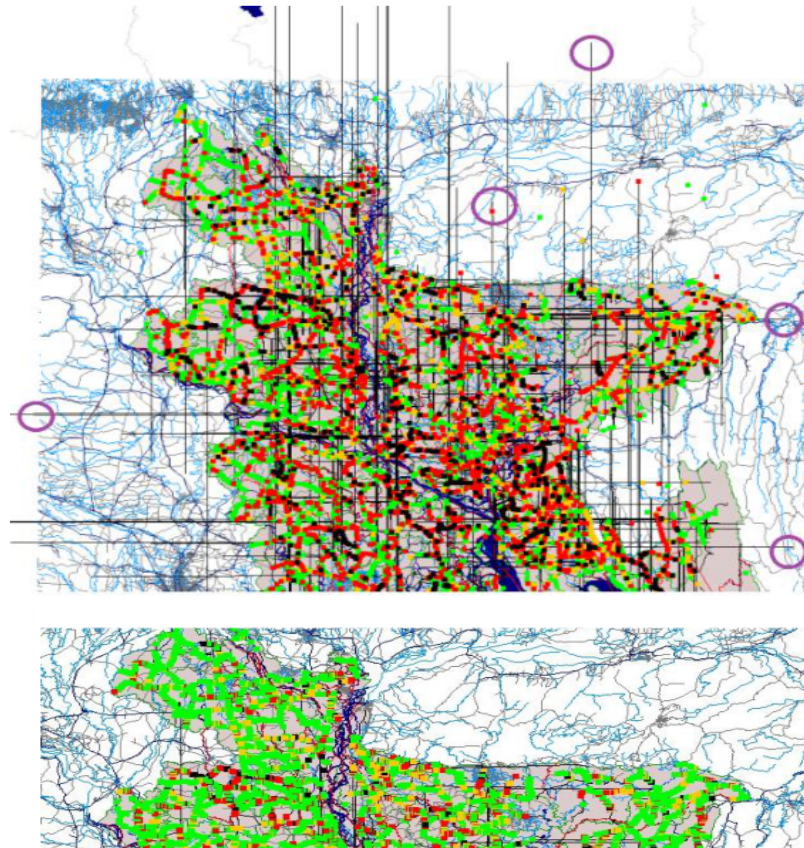


Figure 1. Example of road geometry outliers before and after cleaning.

3. Simulation model

3.1 Model overview

The simulation represents vehicle trips between sources and sinks over the road network. Trips follow shortest paths computed on the graph representation. Bridges are treated as components that can experience stochastic disruption events. When a bridge breaks down, a delay is added to trips that traverse the bridge. In this implementation, disruptions are modeled as delays rather than full link removals, providing a tractable representation of reduced passability and operational disruption.

3.2 Experimental design and scenarios

Scenarios 0-4 represent increasing disruption probabilities for bridges by condition category (A to D). Each scenario is simulated with multiple runs to capture stochastic variation in breakdown realizations.

Scenario	Cat A (%)	Cat B (%)	Cat C (%)	Cat D (%)
0	0	0	0	0
1	0	0	0	5
2	0	0	5	10
3	0	5	10	20
4	5	10	20	40

Table 1. Breakdown probabilities by bridge condition category and scenario.

4. Simulation outcomes

4.1 Driving time response to bridge disruption scenarios

Driving time is used as the primary performance indicator. Figure 2 compares baseline travel time to realizations under the most severe scenario. Figure 3 decomposes outcomes into accumulated delay time and total driving time by scenario. A key empirical pattern is that delay time increases moderately across scenarios, while driving time increases disproportionately.

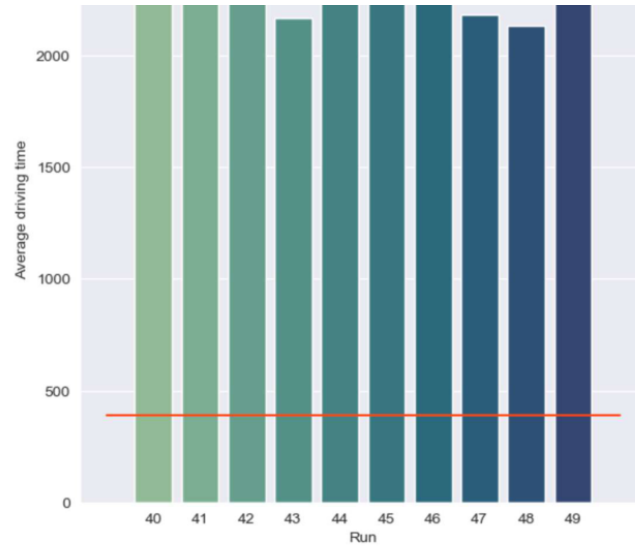


Figure 1: Comparing driving times between baseline and scenario 4. Blue bars depict measured driving time for each run. The first digit in runs refers to the scenario, the second digit refers to the run inside that scenario. The red line depicts driving time in baseline.

Figure 2. Baseline driving time versus Scenario 4 realizations.

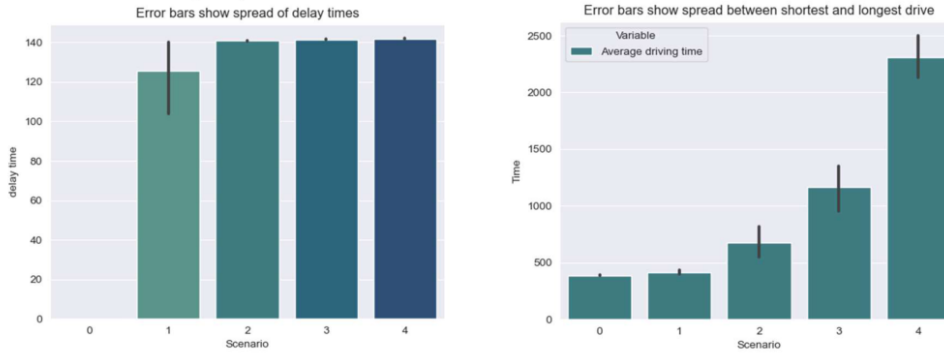


Figure 3. Delay time and total driving time by scenario (mean with range across runs).

Across scenarios 0-4, average driving time increases approximately six-fold from baseline to the most severe scenario. This non-linear response indicates that outcomes are driven not only by the magnitude of delays but also by the network position of affected bridges (e.g., disruptions on high-centrality components disproportionately increase travel time).

4.2 Network structure and critical components

To interpret disruption outcomes and support prioritization, the network was analyzed using centrality measures. Betweenness centrality identifies components that sit on a high share of shortest paths; closeness centrality captures how near a component is to all others in terms of shortest-path distance.

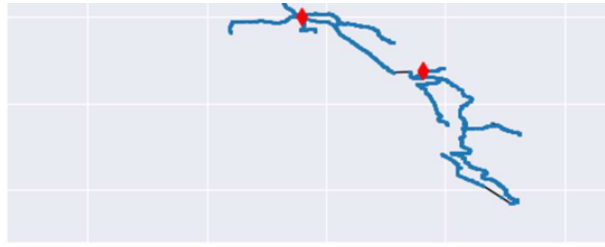


Figure 4. Geographic distribution of highly connected nodes (highest degree).

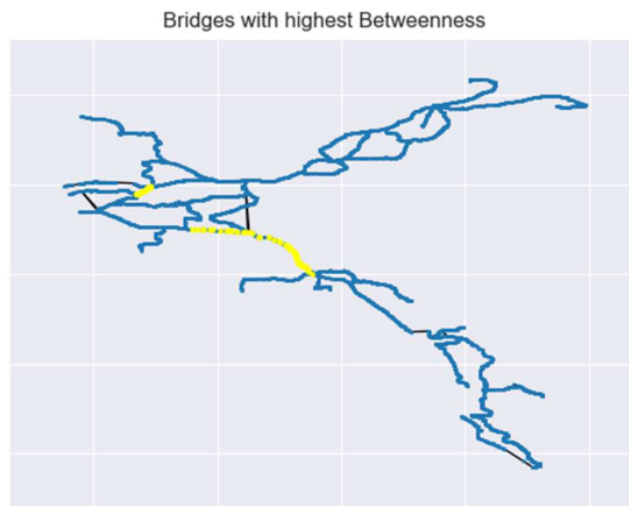


Figure 5. Bridges with highest betweenness centrality highlighted on the network.

4.2.1 Top-ranked intersections and bridges

The following rankings are computed on the modeled N1-N2 subnetwork:

Table 2. Intersections with highest betweenness centrality.

road 1	road 2	id	betweenness centrality
N1	N102	2756	0,513072066
N1	N104	6956	0,496662342
Z1031	Z1034	6956	0,496662342
N2	R211	3141	0,492144605
R211	R310	3141	0,492144605

Table 3. Bridges with highest betweenness centrality.

road	id	condition	bridge name	betweenness centrality
N1	1213	A	AMTOLI SLAB CULVERT	0,485825884

N1	1217	A	DHANPUR RCC GIDER BRIDGE	0,485593787
N1	1229	B	PADUAR BAZER RCCGIDER	0,484886256
N1	1231	C	Mostapur Bridge	0,484766694
N1	1238	A	RAZAPARA RCC GIDER BRIDGE	0,484344541

Table 4. Intersections with highest closeness centrality.

road 1	road 2	id	closeness centrality
N1	Z1042	1158	0,001794412
N1	Z1402	1133	0,001789187
N1	N102	1210	0,001789173
N1	N102	2756	0,00178901
N1	R301	6015	0,001753364

Table 5. Bridges with highest closeness centrality.

road	id	condition	bridge name	closeness centrality
N1	1156	B	ILLIOT BAZER	0,001793967
N1	1163	B	KHAT GHOR BOX CULVERT	0,001793847
N1	1167	B	BAKRA BOX CULVERT	0,001793396
N1	1152	C	AITAL BANGA BOX CULVERT	0,001793077
N1	1173	B	MATIA BAZER	0,001792719

A consistent finding is that many top-ranked components lie on N1, indicating that disruption risk on this corridor can have system-wide effects. Several bridges have betweenness centrality near 0.48-0.49, meaning they sit on nearly half of shortest paths in the modeled subnetwork.

4.3 Criticality and vulnerability metrics

Beyond generic centrality, criticality and vulnerability were operationalized as risk-informed metrics that combine consequence and likelihood. Criticality captures components that simultaneously (i) carry high traffic or sit on many shortest paths and (ii) face higher disruption probability based on condition category. Vulnerability analysis examines how disruption impacts vary by source-sink (origin-destination) structure.

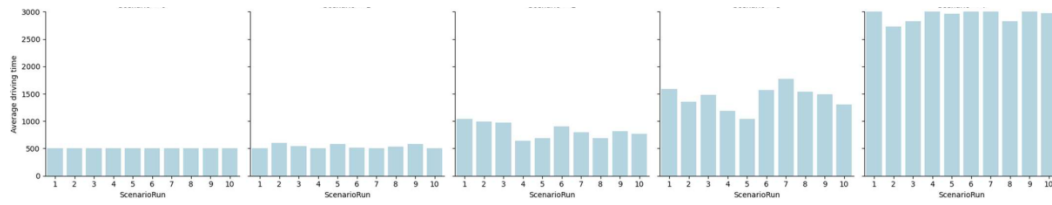


Figure 1: Driving time across scenarios.

Although there are slight differences in driving times between scenarios 1 through 4, they are not as significant as initially anticipated. This is unexpected, given the drastic impact of breakdown probabilities on scenarios. However, upon examining the data in Table 2, it becomes clear what is responsible for this behaviour.

Figure 6. Driving time distributions across scenarios (multiple runs per scenario).

bridges for each source-sink. The determination of vulnerable bridges is based on the analysis of delay frequencies (Figure 2) and delay durations (Figure 3) for each source-sink.

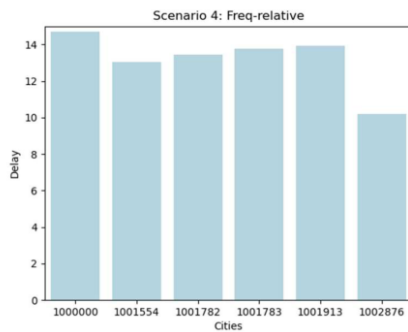


Figure 2: Frequencies of delays across source-sinks in scenario 4.

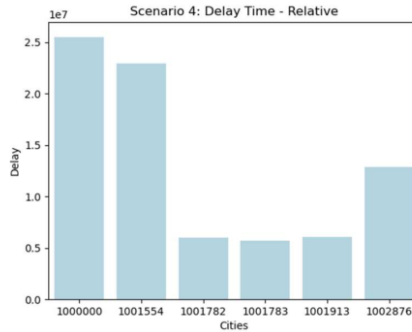


Figure 3: Delay times across source-sinks in scenario 4

Figure 7. Relative delay frequency and delay magnitude by source-sink in Scenario 4.

The source-sink results highlight heterogeneity: some origin-destination groupings experience higher disruption frequency or higher cumulative delay. This supports prioritization that accounts for both network topology and the socio-economic relevance of corridors and hubs (e.g., freight routes, connections to major cities or ports).

5. Implications for policy and investment prioritization

1. Prioritize components that combine high consequence (high betweenness / high traffic) with higher disruption likelihood (poorer condition category).
2. Use network analytics to identify a short list of bridges and junctions on main corridors (especially N1) where targeted strengthening yields disproportionate system benefits.

3. Complement maintenance on critical bottlenecks with redundancy planning: where feasible, increase availability of alternative routes around key bottlenecks to reduce single-point-of-failure risk.
4. Institutionalize a repeatable analytics pipeline (data refresh, scenario re-runs, dashboards) to keep prioritization aligned with changes in asset condition and network use.

6. Limitations and next steps

- Disruptions are modeled as delays rather than full link removals and explicit rerouting; future work could incorporate dynamic route choice and capacity constraints.
- Breakdown probabilities are scenario-based and not calibrated to hazard exposure or observed failure rates; integrating historical hazard and asset condition data would improve realism.
- Traffic demand is simplified; incorporating observed origin-destination matrices, freight flows, and economic weights would strengthen policy relevance.
- Geospatial preprocessing choices (e.g., intersection inference) can affect network topology; systematic validation against authoritative road network layers is recommended.

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